```dart
/*
 Task 1 - Question 1

 Question:
 What is the value of x, y, z and w? (solve it in text file or using comments in dart file)
 int x = 3;
 int y = 2;
 int z = x++;
 int w = ++y;

 --- Step-by-Step Tracing & Explanation ---
 1. int x = 3;
   - Variable 'x' is initialized with the value 3.
   [x = 3]

 2. int y = 2;
   - Variable 'y' is initialized with the value 2.
   [y = 2]

 3. int z = x++;
   - 'x++' is a Postfix Increment operator.
   - The current value of 'x' (3) is assigned to 'z' FIRST.
   - AFTER the assignment, 'x' is incremented by 1 (x becomes 4).
   [z = 3, x = 4]

 4. int w = ++y;
   - '++y' is a Prefix Increment operator.
   - 'y' is incremented by 1 FIRST (y becomes 3).
   - THEN, the new value of 'y' (3) is assigned to 'w'.
   [y = 3, w = 3]

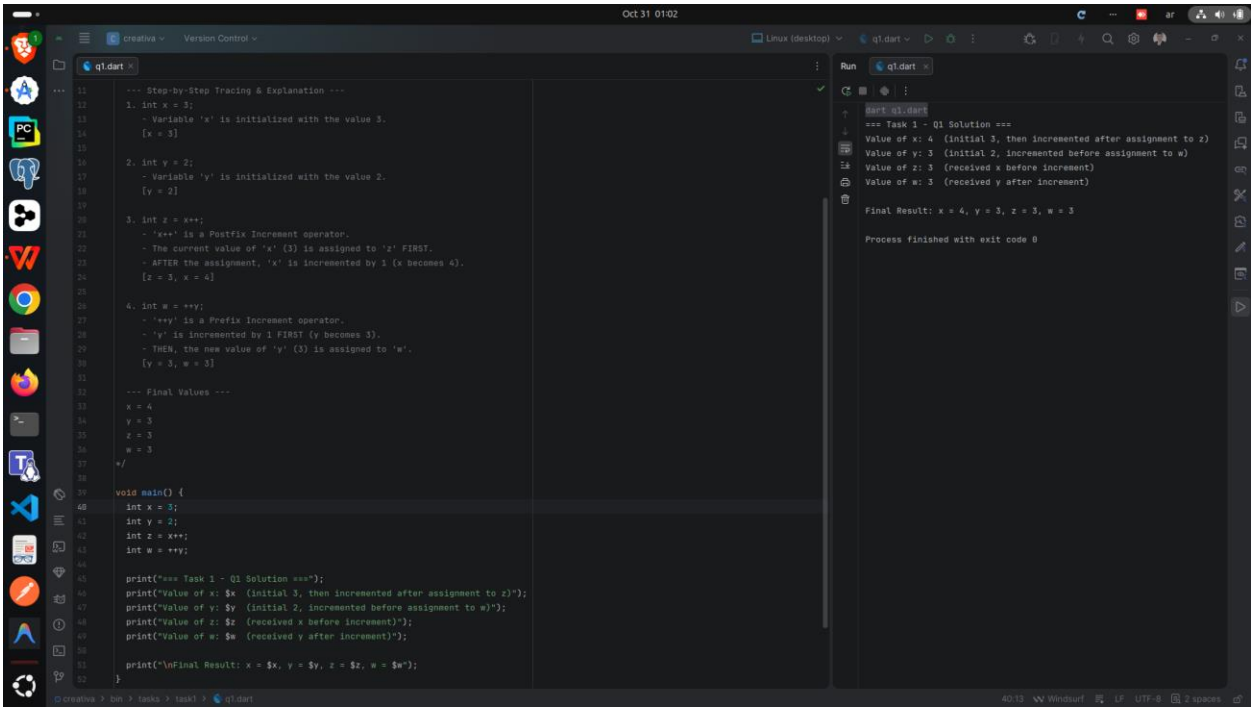
 --- Final Values ---
 x = 4
 y = 3
 z = 3
 w = 3
*/

void main() {
 int x = 3;
 int y = 2;
 int z = x++;
 int w = ++y;

 print("=== Task 1 - Q1 Solution ===");
 print("Value of x: $x  (initial 3, then incremented after assignment to z)");
 print("Value of y: $y  (initial 2, incremented before assignment to w)");
```

```
  print("Value of z: $z  (received x before increment)");
  print("Value of w: $w  (received y after increment)");

  print("\nFinal Result: x = $x, y = $y, z = $z, w = $w");
}
```




```
import 'dart:io';
import 'dart:math';

/*
 Task 1 - Question 2: Compound Interest Calculator (Data Types + Operators)

 Question:
 Calculate the compound interest using the following formula:
 A = P * (1 + r/100)^n
 Where:
 - P = Principal amount
 - r = Annual interest rate (percentage)
 - n = Time period in years
 - A = Total accrued amount
 - Interest Earned = A - P
*/

/// Centralized helper function to prompt and force valid numeric input
double readDouble(String prompt, {bool allowZero = false}) {
```

```
  while (true) {
    stdout.write(prompt);
    double? value = double.tryParse(stdin.readLineSync() ?? "");
    if (value != null && (allowZero ? value >= 0 : value > 0)) {
      return value;
    }
    print("Invalid input! Value is required and must be ${allowZero ? '>= 0' : '> 0'}.");
  }
}

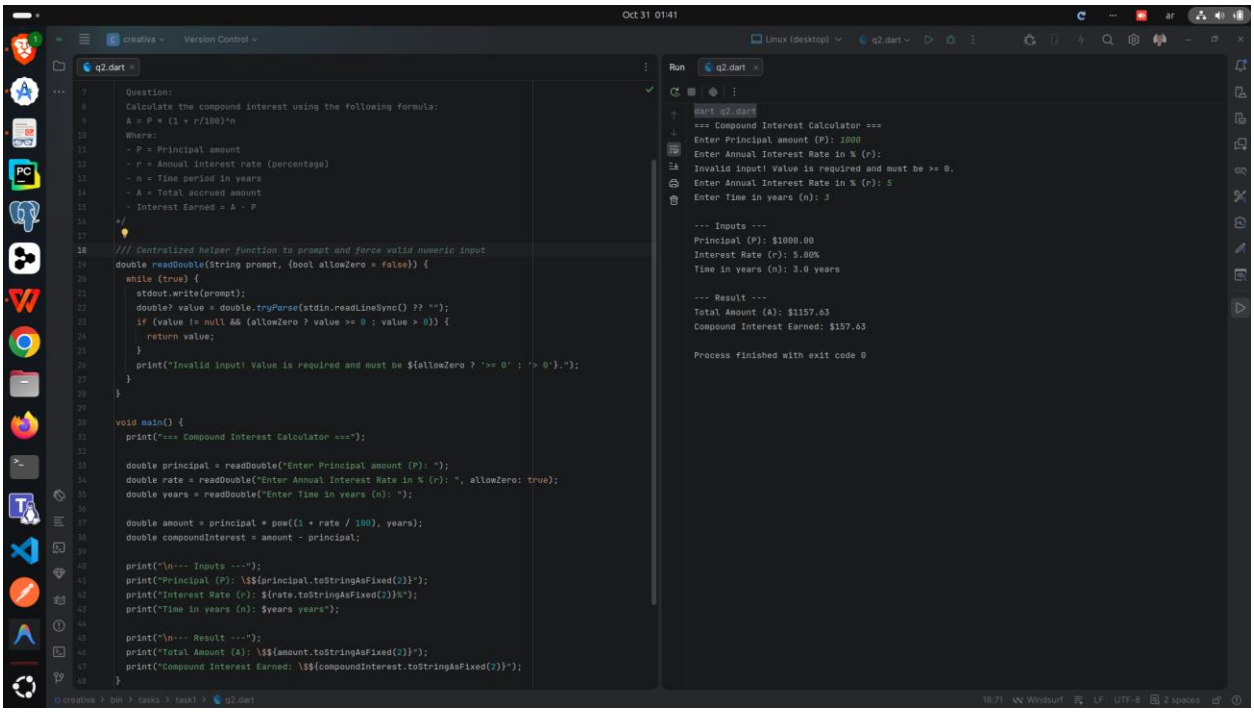
void main() {
  print("=== Compound Interest Calculator ===");

  double principal = readDouble("Enter Principal amount (P): ");
  double rate = readDouble("Enter Annual Interest Rate in % (r): ", allowZero: true);
  double years = readDouble("Enter Time in years (n): ");

  double amount = principal * pow((1 + rate / 100), years);
  double compoundInterest = amount - principal;

  print("\n--- Inputs ---");
  print("Principal (P): \$${principal.toStringAsFixed(2)}");
  print("Interest Rate (r): ${rate.toStringAsFixed(2)}%");
  print("Time in years (n): $years years");

  print("\n--- Result ---");
  print("Total Amount (A): \$${amount.toStringAsFixed(2)}");
  print("Compound Interest Earned: \$${compoundInterest.toStringAsFixed(2)}");
}
```




```dart
import 'dart:io';

/*
 Task 1 - Question 3: Research Question - How to Read User Input in Dart?

 Research & Exploration:
 -----------------------
 In Dart, user input from the console/terminal is handled via the `dart:io` library,
 specifically using the standard input stream `stdin`.

 Key Ways to Read Input:
 -----------------------
 1. Synchronous Line Reading: `stdin.readLineSync()`
   - The most common method. Reads a line from console until Enter (newline) is pressed.
   - Returns a nullable String (`String?`), which is null on EOF (End-Of-File).
   - Can be converted to numbers using:
     * int.parse(input!) / int.tryParse(input ?? "")
     * double.parse(input!) / double.tryParse(input ?? "")

 2. Synchronous Single Byte Reading: `stdin.readByteSync()`
   - Reads a single byte / keypress code from standard input synchronously.
   - Returns an integer byte code (-1 if end of stream).

 3. Asynchronous Reading via Stream (`stdin.transform(...)`):
   - `stdin` implements `Stream<List<int>>`.
   - In asynchronous or server applications, you can listen to lines:
```

```
    `stdin.transform(utf8.decoder).transform(const LineSplitter()).listen((line) { ... });`

  4. Command-Line Arguments: `main(List<String> args)`
    - Data can also be passed when executing the script:
     `dart run bin/tasks/task1/q3.dart arg1 arg2`
*/

void main(List<String> args) {
  print("=== Research Question: Reading User Input in Dart ===\n");

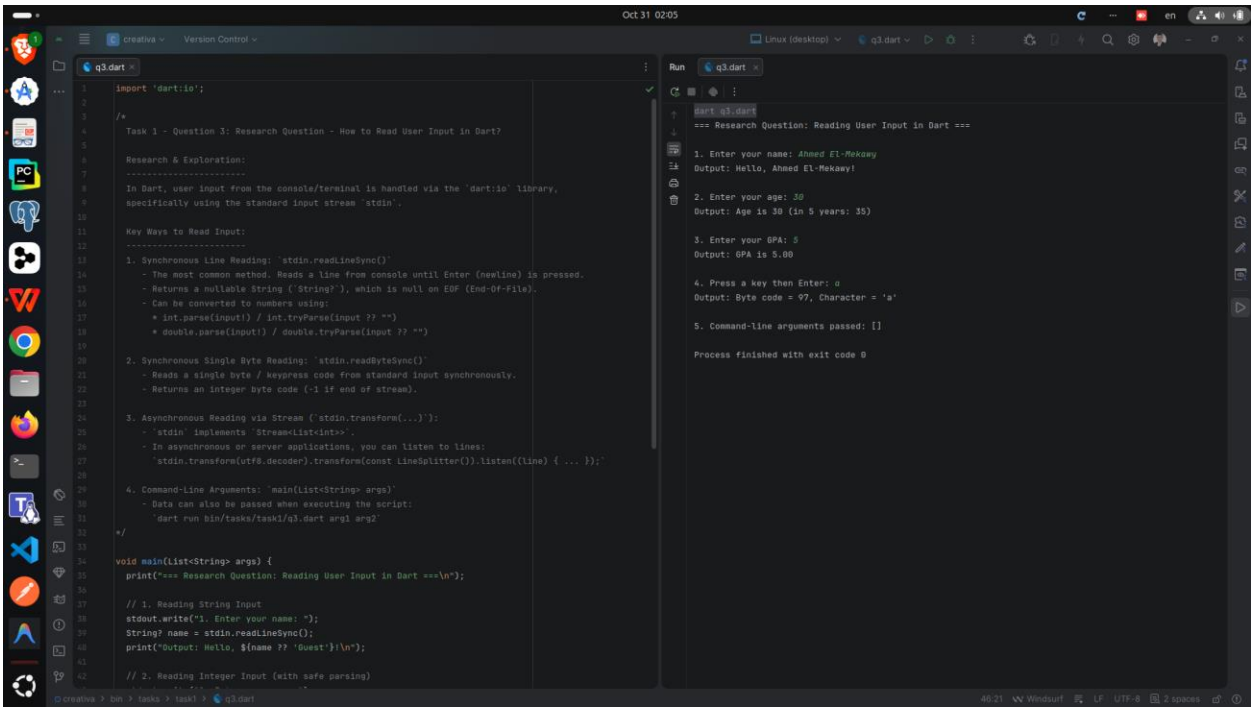
  // 1. Reading String Input
  stdout.write("1. Enter your name: ");
  String? name = stdin.readLineSync();
  print("Output: Hello, ${name ?? 'Guest'}!\n");

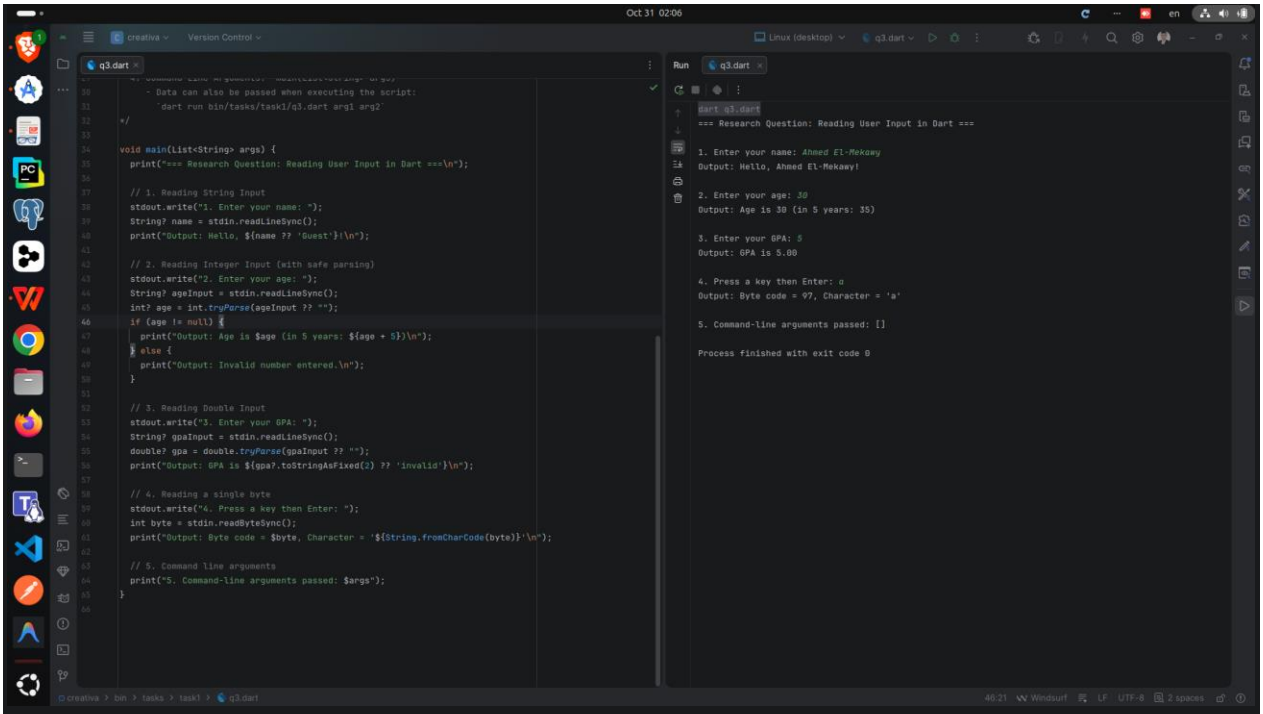
  // 2. Reading Integer Input (with safe parsing)
  stdout.write("2. Enter your age: ");
  String? ageInput = stdin.readLineSync();
  int? age = int.tryParse(ageInput ?? "");
  if (age != null) {
    print("Output: Age is $age (in 5 years: ${age + 5})\n");
  } else {
    print("Output: Invalid number entered.\n");
  }

  // 3. Reading Double Input
  stdout.write("3. Enter your GPA: ");
  String? gpaInput = stdin.readLineSync();
  double? gpa = double.tryParse(gpaInput ?? "");
  print("Output: GPA is ${gpa?.toStringAsFixed(2) ?? 'invalid'}\n");

  // 4. Reading a single byte
  stdout.write("4. Press a key then Enter: ");
  int byte = stdin.readByteSync();
  print("Output: Byte code = $byte, Character = '${String.fromCharCode(byte)}'\n");

  // 5. Command line arguments
  print("5. Command-line arguments passed: $args");
}
```






```dart
import 'dart:io';

/*
 Task 1 - Question 4: Distance & Speed Calculator (Arithmetic Operators + Data Types)

 Question:
 Calculate the time needed to travel a certain distance at a given speed,
```

```dart
 then convert the time from hours to minutes and seconds.

 Formulas:
 - timeInHours = distance / speed
 - totalMinutes = timeInHours * 60
 - totalSeconds = timeInHours * 3600
*/

/// Centralized helper function to prompt and force valid numeric input
double readDouble(String prompt, {bool allowZero = false}) {
 while (true) {
  stdout.write(prompt);
  double? value = double.tryParse(stdin.readLineSync() ?? "");
  if (value != null && (allowZero ? value >= 0 : value > 0)) {
   return value;
  }
  print("Invalid input! Value is required and must be ${allowZero ? '>= 0' : '> 0'}.");
 }
}

void main() {
 print("=== Distance & Speed Calculator ===");

 double distance = readDouble("Enter distance (e.g. in km or miles): ");
 double speed = readDouble("Enter speed (e.g. in km/h or mph): ");

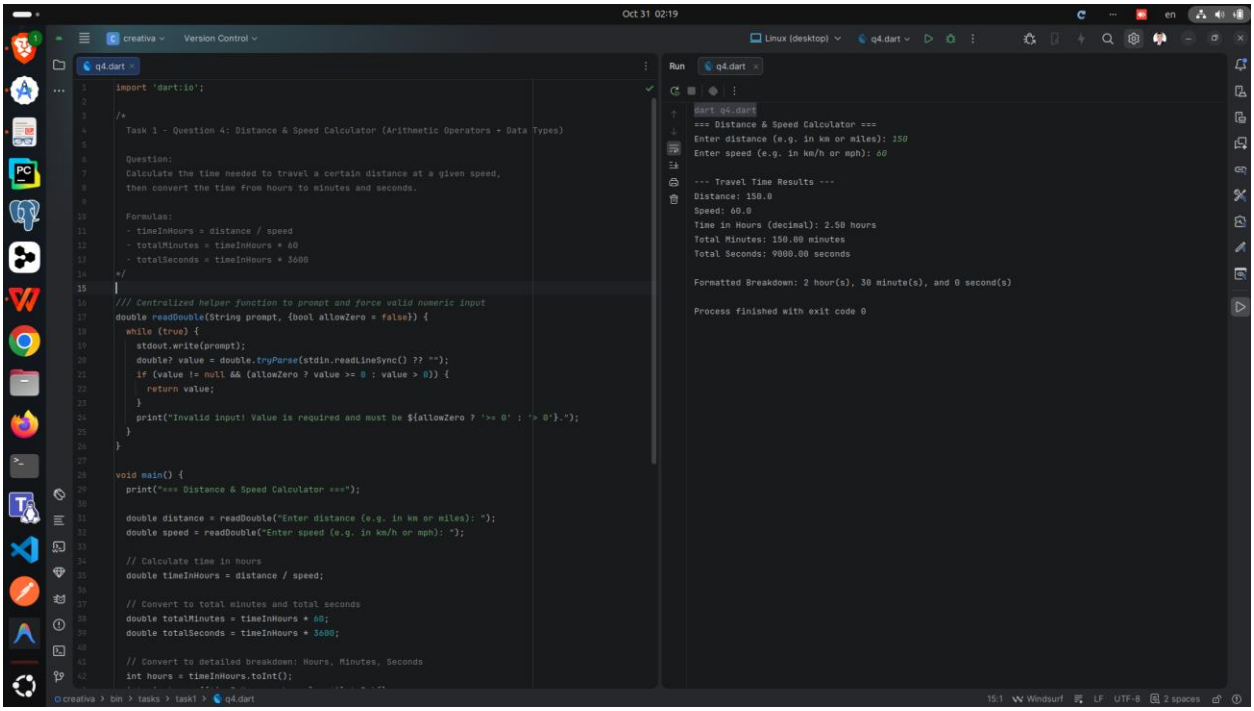
 // Calculate time in hours
 double timeInHours = distance / speed;

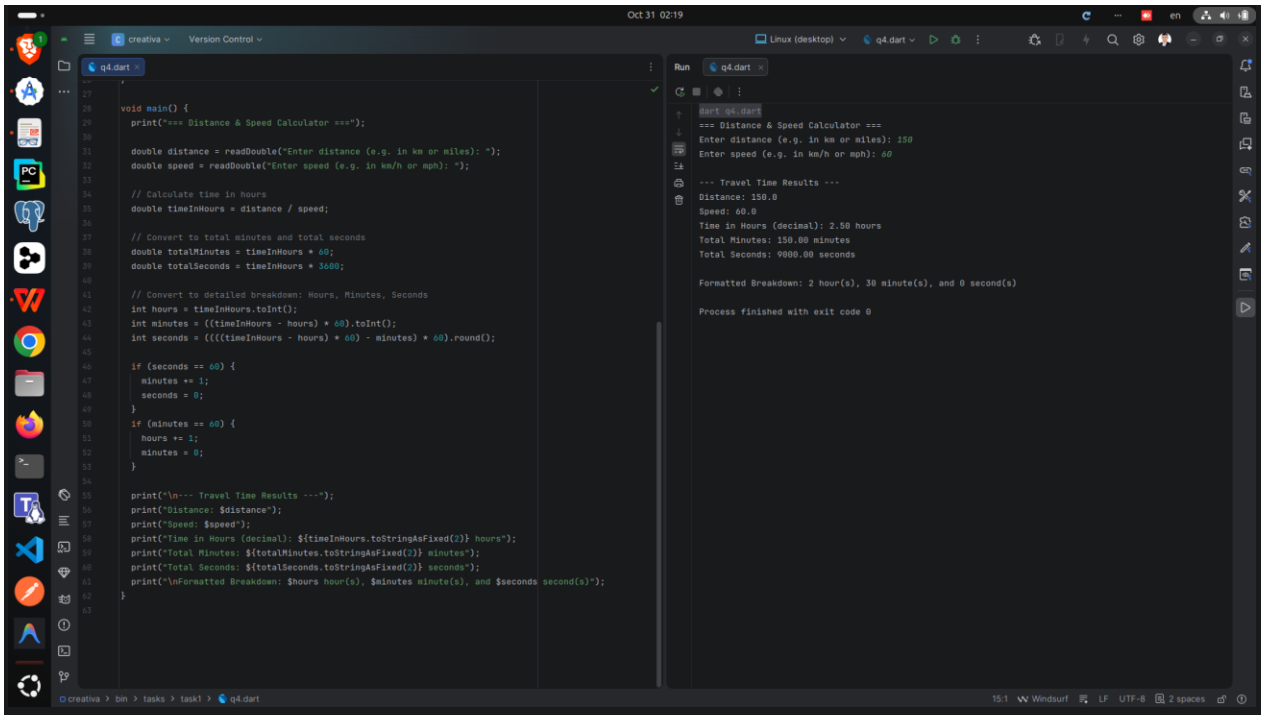
 // Convert to total minutes and total seconds
 double totalMinutes = timeInHours * 60;
 double totalSeconds = timeInHours * 3600;

 // Convert to detailed breakdown: Hours, Minutes, Seconds
 int hours = timeInHours.toInt();
 int minutes = ((timeInHours - hours) * 60).toInt();
 int seconds = ((((timeInHours - hours) * 60) - minutes) * 60).round();

 if (seconds == 60) {
  minutes += 1;
  seconds = 0;
 }
 if (minutes == 60) {
  hours += 1;
  minutes = 0;
 }
```

```
  print("\n--- Travel Time Results ---");
  print("Distance: $distance");
  print("Speed: $speed");
  print("Time in Hours (decimal): ${timeInHours.toStringAsFixed(2)} hours");
  print("Total Minutes: ${totalMinutes.toStringAsFixed(2)} minutes");
  print("Total Seconds: ${totalSeconds.toStringAsFixed(2)} seconds");
  print("\nFormatted Breakdown: $hours hour(s), $minutes minute(s), and $seconds second(s)");
}
```






```dart
import 'dart:io';
import 'dart:math';

/*
 Task 1 - Question 5 (Bonus+): Compute the volume of a cylinder

 Question:
 Write a program that reads in the radius and length of a cylinder
 and computes the area and volume using the following formulas:
 - area = radius * radius * pi
 - volume = area * length
*/

/// Centralized helper function to prompt and force valid numeric input
double readDouble(String prompt, {bool allowZero = false}) {
 while (true) {
  stdout.write(prompt);
  double? value = double.tryParse(stdin.readLineSync() ?? "");
  if (value != null && (allowZero ? value >= 0 : value > 0)) {
   return value;
  }
  print("Invalid input! Value is required and must be ${allowZero ? '>= 0' : '> 0'}.");
 }
}

void main() {
 print("=== Cylinder Area & Volume Calculator ===");
```

```
  double radius = readDouble("Enter the radius of the cylinder: ");
  double length = readDouble("Enter the length of the cylinder: ");

  // Calculate base area and volume
  double area = radius * radius * pi;
  double volume = area * length;

  print("\n--- Results ---");
  print("Radius: $radius");
  print("Length: $length");
  print("Base Area: ${area.toStringAsFixed(4)}");
  print("Volume: ${volume.toStringAsFixed(4)}");
}
```

